



Unveiling the Potential of Biogenic Nanoparticles for Wastewater Remediation: A Systematic Review

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Abstract In recent decades, escalating water pollution driven by demographic growth, industrial development, agricultural intensification, and rapid urban expansion has emerged as a critical global environmental challenge. Contamination of water bodies by heavy metals, organic compounds, and pathogenic microorganisms has intensified the need for sustainable wastewater remediation strategies. This review elucidates the mechanistic potential and strategies for wastewater remediation using biogenic nanoparticles (BNPs) synthesized via plant, bacterial, fungal, and algal systems as emerging multifunctional materials for addressing complex wastewater matrices. Conventional wastewater treatment methods, such as coagulation, flocculation, and sedimentation, have limitations in efficiency, chemical demand, and operational costs, prompting interest in innovative BNP-based approaches. BNPs demonstrated enhanced adsorption, catalytic transformation, and antimicrobial activity due to their biomolecule-rich surfaces, enabling efficient contaminant removal across industrial,

municipal, textile, pharmaceutical, and agricultural wastewater systems. It has been evaluated BNP synthesis routes and mechanisms, pollutant removal efficiencies, environmental behavior, and implementation challenges. This review summarized that BNPs achieve enhanced adsorption, catalytic transformation, and antimicrobial activity, with removal efficiencies of 80–98% for dyes and 70–95% for heavy metals, while offering improved biocompatibility and reduced ecotoxicity compared with chemically synthesized nanoparticles. This review also highlights critical barriers to large-scale application, including synthesis variability, economic scalability, regulatory uncertainty, and community acceptance. Moreover, environmental fate analyses reveal that post-treatment transformations, including sulfidation, organometallic complexation, and surface passivation, play a decisive role in governing BNPs' persistence, bioavailability, and ecological impact.

Keywords Nanoremediation · Wastewater · Contamination · Pollutant removal · Ecotoxicity · Biogenic · Treatment

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1 Introduction

Water pollution is a major challenge worldwide. Approximately 5.9 billion people lack access to clean water for basic hygiene, and 2 billion lack access to clean drinking water (Hameed et al., 2025; Liu

et al., 2025a, b), as surface and groundwater quality declines. Beyond surface water depletion, groundwater contamination is a co-existing global challenge driven by uncontrolled industrial discharge, agricultural runoff, and infiltration of untreated effluents, which contribute to accumulation of heavy metals, nitrates, and emerging organic pollutants in riverine systems (Yu et al., 2025). Most people facing clean water scarcity in Asia and Africa. Clean water is necessary not only for human consumption but also for pharmaceutical industry, food processing, and the medical sectors (Tudu et al., 2023).

Despite its crucial role in sustaining life, surface water is highly susceptible to pollution from intensive industrialization and uncontrolled agricultural practices. Approximately 90% of untreated sewage in developing nations is discharged directly into water bodies (Tariq et al., 2023). Wastewater contains a diverse contaminants that can be broadly classified into chemical, biological, persistent organic pollutants (POPs), and emerging contaminants (Mateescu et al., 2024). Chemical pollutants include heavy metals (e.g., Pb, Cd, Cr), inorganic ions, and industrial solvents; biological contaminants include pathogenic bacteria, viruses, protozoa, and helminths, all of which pose direct public health risks. Persistent Organic Pollutants, such as polycyclic aromatic hydrocarbons (PAHs) and chlorinated pesticides, are highly stable, bioaccumulative, and toxic. Emerging contaminants, including pharmaceuticals, endocrine-disrupting compounds (EDCs), and volatile organic compounds (VOCs), have gained global attention due to their persistence and ecological effects (Zhang et al., 2023).

Contamination of water bodies by synthetic dyes and other industrial colorants alters key water quality parameters, including chemical oxygen demand (COD), biological oxygen demand (BOD), salinity, and light penetration, thereby affecting photosynthetic activity, aquatic biodiversity, and human health (Kulkarni et al., 2024). Adding synthetic dyes alters chemical and biological oxygen demand, salt concentration, light penetration, and water appearance, ultimately harming aquatic and human health. Moreover, even at low doses, heavy metals cause multiple organ failure, cancer, neurological problems, and mutations (Hairom et al., 2021). This study focuses on remediating chemical and persistent organic pollutants, including heavy metals, dyes, PAHs, VOCs,

and selected emerging contaminants, using biogenic nanoparticles as sustainable nanoremediation agents.

As new pollutants emerge, traditional wastewater treatment strategies are no longer efficient. Rapid urbanization has led to high concentrations of pollutants, including heavy metals, organic & inorganic pollutants, toxic dyes, and pathogenic microorganisms, in wastewater. When wastewater is properly treated, it can be reused for various freshwater-demanding purposes (Xiao et al., 2017). Over the past few decades, conventional water treatment techniques, including sedimentation, flocculation, and coagulation, have consistently shown high efficiency in removing pollutants, ensuring effluent quality suitable for discharge or reuse. Furthermore, inadequate removal of emerging contaminants necessitates rigorous advancements and exploration of alternative approaches (Gedda et al., 2021). Therefore, using nanoparticles to remove pollutants from wastewater is an interesting and emerging area of research (Ali et al., 2020).

1.1 Economic Considerations and Cost Challenges in Nanotechnology-Based Wastewater Treatment

While nanotechnology-based wastewater treatment is widely regarded as a cost-effective and efficient alternative to conventional methods, its overall economic feasibility remains context-dependent. The synthesis, stabilization, and large-scale deployment of nanoparticles, particularly biogenic variants, entail high costs for precursor materials, energy inputs, and recovery processes (Velamakanni et al., 2026). According to Zhang et al. (2025), the estimated production cost of high-purity nanoparticles used for wastewater treatment ranges from USD 180 to 300 per gram, primarily influenced by precursor selection, biological source, and synthesis efficiency. Moreover, challenges in nanoparticle regeneration, reusability, and separation from treated effluents further increase operational expenses, often raising total treatment costs by 20–35% relative to conventional adsorption methods. Similarly, Ahmed et al. (2021) noted that while green and biogenic synthesis routes reduce the need for hazardous chemicals and lower environmental costs, it still require optimized process control, microbial culture maintenance, and waste biomass management, which can offset some anticipated economic benefits. However, as scalable biosynthesis and nanoparticle

recovery technologies advance, production costs are expected to decline significantly in the coming years (Damian et al., 2025). Therefore, while nanotechnology offers superior treatment performance and long-term sustainability, optimizing costs and lifecycle economics remains critical for large-scale implementation, particularly in developing regions (Khan et al., 2025a, b).

1.2 Nanotechnology as a Foundation for Advanced Pollutant Removal Strategies

Nanotechnology encompasses the synthesis and application of a diverse range of nanomaterials, including structures between 1 and 100 nm that exhibit unique optical, catalytic, and physicochemical properties not found in their bulk counterparts (Mekuye et al., 2023). Biogenic nanoparticles are synthesized using plants, bacteria, fungi, and algae. Nanotechnology can help to reduce and stabilize the nanoparticles, therefore reducing the use of toxic chemicals. During the process, substances such as proteins, enzymes, sugars, and other naturally produced chemicals help turn metal ions into nano-sized particles. It also reduce aggregation and help maintain the nanostructure (Mughal et al., 2021). Evaluating nanoparticles in environmental matrices and the resulting residues is essential for assessing their efficacy, stability, and potential ecological impact during wastewater treatment. Standard testing approaches quantify removal performance using parameters such as adsorption capacity, removal efficiency, and residual pollutant concentration, typically determined by techniques such as atomic absorption spectroscopy (AAS), inductively coupled plasma mass spectrometry (ICP-MS), and UV–Vis spectrophotometry (Bersier et al., 1994).

In addition, physicochemical characterization of nanoparticles before and after treatment uses methods such as dynamic light scattering (DLS) for particle size distribution, zeta potential analysis for surface charge and colloidal stability, and electron microscopy (SEM/TEM) for morphological assessment (Wadhwa et al., 2022). Leaching studies, stability assessments, and toxicity assays (e.g., microbial inhibition or ecotoxicity tests) further evaluate nanoparticle fate and transformation in treated water and sludge residues, helping minimize secondary environmental risks. Collectively, these analytical and monitoring strategies provide a comprehensive framework

for assessing nanoparticle behavior, treatment performance, and environmental safety in wastewater remediation systems (Abdulsada et al., 2021).

Nanomaterials are widely used in biomedicine, pharmaceuticals, electronics, environmental detection, cosmetics, and textile industries (Ilangovan et al., 2021). In wastewater treatment, their exceptionally high surface area-to-volume ratio enhances photocatalytic activity, increases adsorption capacity, and enables rapid pollutant removal. Thus, nano-adsorbents and nanocatalysts are widely used to remediate organic and inorganic contaminants; for instance, cobalt oxide nanoparticles for dye degradation and metal oxide or carbon-based nanoparticles for heavy metal removal (Dehghani et al., 2019; Rashid et al., 2021).

In recent years, research attention has also increased on biogenic nanoparticles, a subclass of nanomaterials synthesized via biological routes using plants, bacteria, fungi, algae, or yeast (Bokolia et al., 2025). These BNPs are generated from naturally occurring biomolecules that serve as reducing, stabilizing, and capping agents, thereby imparting biocompatibility and unique surface functionalities. Their biologically derived coatings enhance selectivity, catalytic efficiency, and environmental safety, making them particularly relevant for wastewater management, treatment facilities, and soil amendment authorities (Sharma et al., 2024). Introducing BNPs into this framework offers a sustainable alternative to chemically synthesized nanoparticles because of their eco-friendly synthesis, reduced toxicity, and use of renewable biological resources, while enabling a deeper understanding of their mechanisms and applications in advanced wastewater remediation (Nain et al., 2025).

Nanoparticles such as carbon nanotubes, zero-valent nanoparticles, metal oxide nanoparticles, and nanocomposites have been widely used in wastewater remediation because of their high surface reactivity and tunable physicochemical properties (Rezania et al., 2024). Researchers quantitatively assess their effectiveness using parameters such as adsorption capacity, removal efficiency, reaction kinetics, and residual pollutant concentrations, which collectively guide their integration into advanced treatment systems, including membranes, functionalized adsorbents, and catalytic materials (El-Sayed et al., 2020). Nanoparticles enhance the adsorptive and controlled properties of certain materials for specific applications. This provides a variety of options for

remediating contaminated water, including highly reactive, magnetic, flexible, size-dependent, and biodegradable nanostructured materials (Azeez et al., 2023). Nanostructured materials possess unique physicochemical properties that significantly enhance their environmental remediation potential. Compared with conventional adsorbents, nanomaterials exhibit a high surface-area-to-volume ratio, tunable surface charge, and diverse pore morphologies, thereby enabling superior adsorption capacity, rapid diffusion kinetics, and enhanced photocatalytic activity (Ahmaruzzaman et al., 2025). These nanoscale properties enable superior adsorption capacity, rapid diffusion kinetics, and enhanced photocatalytic activity, resulting in efficient pollutant interactions through mechanisms such as electrostatic attraction, redox reactions, and catalytic degradation (Cai et al., 2025). Furthermore, their flexibility in surface functionalization enables selective binding of contaminants, including dyes, heavy metals, and emerging organic pollutants, while maintaining structural stability and reusability under varying environmental conditions (Ahmed & Mubarak, 2021). Collectively, these attributes make nanostructured materials highly effective candidates for sustainable wastewater treatment.

Therefore, this review investigates the potential implications of BNPs for wastewater treatment and pollutant removal, including organic pollutants. Furthermore, the objectives of this review are (a) challenges with respect to conventional treatment methods, (b) multifaceted decontamination strategies, (c) fate and barriers to BNPs in water treatment, and (d) regulatory frameworks for environmental risk mitigation. This work advances water technologies, focusing on environmentally friendly remediation and sustainability.

The novelty of the present review lies in its comprehensive, multidisciplinary synthesis of recent advances in biogenic nanoparticle-mediated wastewater remediation. Unlike earlier reviews that primarily emphasized metal- or plant-derived nanoparticles, this work provides holistic coverage of biogenic systems, encompassing microbial-, fungal-, algal-, and plant-mediated nanoparticles, and elucidates their collective mechanisms for the removal of heavy metals, dyes, organic pollutants, and pathogenic microorganisms. Furthermore, the review provides mechanistic insights across three distinct nanoscale strategies like nanoadsorbents, nanocatalysts, and nanomembranes,

emphasizing how biogenic synthesis influences their physicochemical properties, surface reactivity, and pollutant removal efficiencies. This comparative mechanistic evaluation remains underexplored in the existing literature. Another distinctive contribution is its critical evaluation of the post-treatment fate, transformation, and potential ecotoxicological impacts of biogenic nanoparticles in aquatic environments, supported by recent quantitative studies (Morales et al., 2022; Mukhopadhyay et al., 2022). Additionally, the manuscript assesses emerging global regulatory frameworks and environmental risk mitigation policies, with particular attention to European Union and Indian perspectives, addressing an area often overlooked in prior reviews. Finally, this review follows a PRISMA-guided systematic approach and synthesizes evidence from peer-reviewed studies (2019–2025). Together, these features make this work a novel and comprehensive contribution to sustainable wastewater nanoremediation.

2 Literature Review: Methodology

2.1 Search Strategy and Databases Used

A systematic literature search was conducted on June 26, 2025, in accordance with the PRISMA 2020 guidelines to ensure transparency, reproducibility, and scientific integrity. The search was conducted exclusively using peer-reviewed, citation-indexed databases, Web of Science (WoS) and Scopus, to maintain academic rigor and source reliability. Search strings combined Boolean operators and specific keywords, including: “biogenic nanomaterials,” “green synthesis,” “biogenic nanoparticles,” “nanotechnology,” “wastewater remediation,” and “water treatment.” The search was restricted to peer-reviewed articles published between 2019 and 2025 and written in English. The reference lists of key papers were also manually screened to ensure inclusion of any additional relevant studies. Clearly defined inclusion and exclusion criteria guided study selection, ensuring methodological transparency and academic rigor. Only peer-reviewed, English-language research and review articles published between 2019 and 2025 were included. Studies were further filtered through explicit inclusion criteria: (i) use of plant, microbial, fungal, or algal systems for nanoparticle synthesis; (ii)

evaluation of BNP performance in removing heavy metals, dyes, organic pollutants, pharmaceuticals, or pathogens; and (iii) mechanistic or application-specific insights relevant to environmental remediation. Exclusion criteria eliminated studies focused solely on chemical synthesis, nanoparticle characterization without application, or unrelated nanomaterial types. Each full-text article was critically evaluated based on methodological rigor, experimental relevance, and contribution to the central objective of this review. Eligible studies were required to provide quantitative data, mechanistic explanations, or comparative analyses of pollutant removal efficiency using biogenic nanomaterials derived from microbial, algal, fungal, or plant sources. Publications written in English and containing sufficient experimental or analytical detail to enable critical evaluation were prioritized.

To maintain scientific reliability, non-peer-reviewed materials, such as conference proceedings, theses, reports, and editorials, were excluded. Studies that addressed only chemically or physically synthesized nanoparticles, without any biogenic component, were also excluded, as were papers lacking clearly described methodologies, quantitative outcomes, or pollutant-specific data relevant to wastewater remediation. Duplicate records and publications outside the predefined time frame were removed during screening. This systematic inclusion–exclusion framework ensured that only high-quality, methodologically sound, and contextually relevant studies were included in the final synthesis, thereby enhancing the validity and reproducibility of the review’s findings.

2.2 Data Collection Procedure

The articles obtained from the literature search were selected based on their topics of interest, including conventional wastewater treatment techniques and the role of nanoparticles as nanoadsorbents, nanocatalysts, and nanomembranes. Additionally, various decontamination methods, including the removal of heavy metals, organic pollutants, and dyes, as well as the elimination of pathogenic organisms, were studied. This systematic review followed the PRISMA 2020 guidelines to ensure transparency, reproducibility, and academic reliability. The literature search was conducted exclusively in two highly credible, peer-reviewed, citation-indexed databases: Web of Science (WoS) and Scopus, to ensure inclusion of only

high-quality scientific studies. The search was conducted on June 26, 2025, and included publications from January 2019 to June 2025. A combination of Boolean operators and specific keywords was applied to identify relevant studies, including: “green synthesis,” “biogenic nanoparticles,” “biogenic nanomaterials,” “nanotechnology,” “wastewater treatment,” and “water remediation.”

During the identification phase, a total of 779 studies were retrieved, 412 from Web of Science and 367 from Scopus. Out of all, 128 duplicate records were identified and removed, leaving 651 unique articles for initial screening. In the screening phase, the titles and abstracts of these 651 records were reviewed to determine their relevance to biogenic nanoparticle applications in wastewater treatment. At this stage, 509 articles were excluded due to irrelevance, duplication of concepts, or lack of a clear connection to the review’s focus area. The remaining 142 articles proceeded to the eligibility assessment phase. During the eligibility phase, we evaluated each of the 142 full-text articles against predefined inclusion and exclusion criteria, including experimental design, methodological soundness, data reliability, and direct relevance to biogenic nanotechnology-based wastewater remediation. After this analysis, 70 studies were excluded for inadequate methodological detail, lack of quantitative data, limited focus on biogenic processes, or insufficient reproducibility. Finally, 72 studies met all eligibility criteria and were included in the qualitative synthesis of this review. These studies collectively represent the most recent and methodologically robust advances in biogenic nanomaterial synthesis, characterization, and application in wastewater remediation. Figure 1 illustrates the entire stepwise selection process, including identification, screening, eligibility assessment, and inclusion.

3 Conventional Techniques for Wastewater Remediation

The Industrial Revolution in the late eighteenth century led to the first scientific approach to wastewater treatment, using septic tanks (Arden et al., 1914). The twentieth century saw significant advancements in wastewater remediation, including sedimentation, flocculation, coagulation, chemical precipitation, and trickling filters (Saxena et al., 2025). The

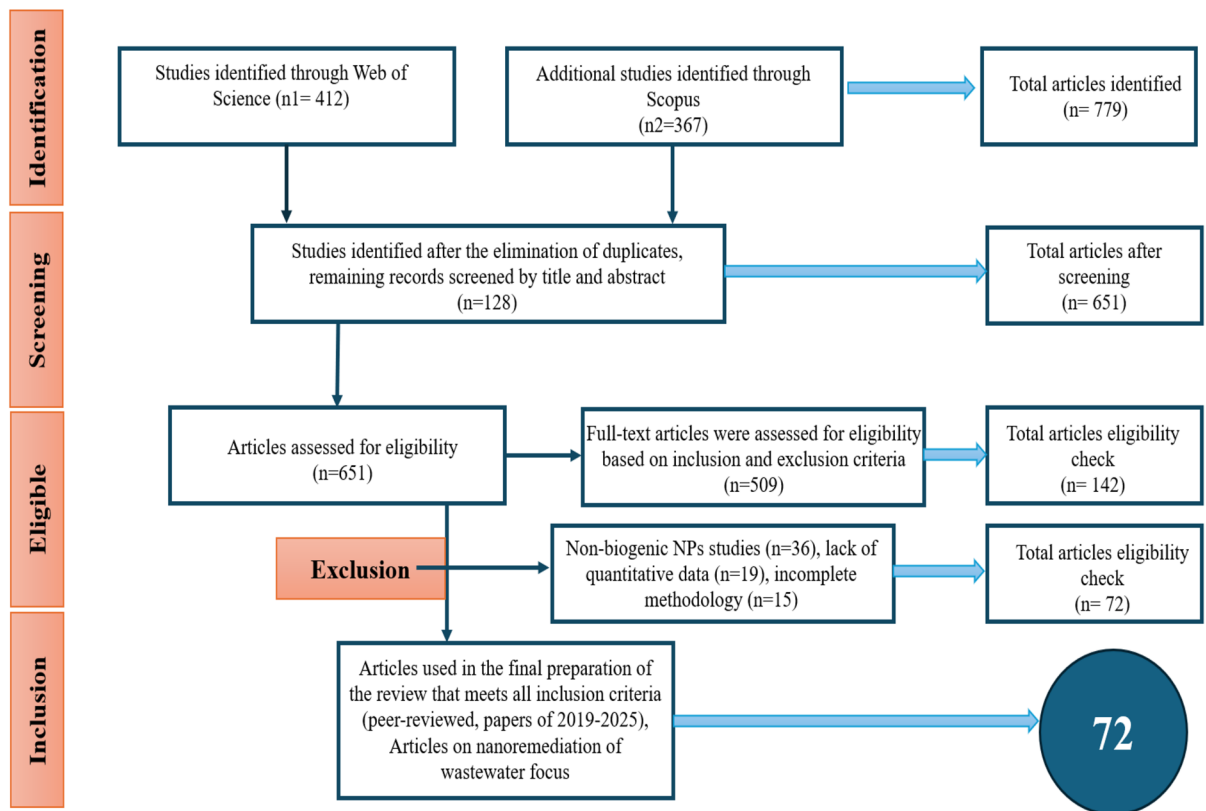


Fig. 1 The PRISMA Flow Diagram: The PRISMA diagram illustrates the systematic workflow used in selecting studies for the review. From 779 initial records identified in Web of Science and Scopus, 128 duplicates were removed, and 651 stud-

ies were screened. Following a detailed eligibility assessment, 72 studies met the inclusion criteria and were incorporated into the final synthesis

conventional treatment processes highlighted in this review- coagulation, flocculation, sedimentation, chemical precipitation, and the biological trickling filter- were selected because it represent the most widely adopted and foundational methods used in municipal and industrial wastewater treatment systems worldwide (Aragaw et al., 2023). The trickling filter, in particular, remains one of the oldest and most widely implemented biological processes because of its low energy requirement, operational simplicity, and established performance in removing biodegradable organic matter (Abyar & Nowrouzi, 2023). To address operational limitations such as sludge generation, poor trace contaminant removal, and long retention times, advanced approaches are needed. Nanobioremediation offers a more efficient alternative with enhanced reactivity and treatment performance (Selvinsimpson & Chen, 2023).

3.1 Sedimentation

Sedimentation is one of the oldest and most fundamental physical treatment processes used for wastewater clarification. It primarily relies on the gravitational settling of suspended particles and colloids to separate solids from liquid effluents (Peters et al., 1984). Although it remains a critical pretreatment step in most wastewater treatment plants, traditional sedimentation is often inefficient at removing fine particulates and dissolved pollutants, and it requires large spaces and long retention times (Rezai et al., 2021). Traditional sedimentation is often inefficient at removing fine particulates and dissolved pollutants, requiring large spaces and long retention times, although it remains a critical pretreatment step in most wastewater treatment plants. Recent studies,

however, have re-evaluated its role within integrated hybrid systems, where it works synergistically with advanced technologies such as membrane filtration and adsorption (Najem et al., 2024). According to Jiang et al. (2025), conventional sedimentation alone achieves only 45–55% removal efficiency for suspended solids and turbidity, whereas integration with nanomembrane filtration can enhance removal efficiency up to 92–95%, significantly improving effluent quality. Similarly, Mubarak et al. (2022) reported that coupling sedimentation with adsorption-based systems using biogenic nanomaterials increased pollutant removal by nearly 40% compared to stand-alone sedimentation. These improvements are attributed to enhanced surface interactions and reduced particle aggregation facilitated by nanostructured adsorbents. Furthermore, advanced polymer-assisted sedimentation processes have demonstrated notable performance gains. El-Sheikh et al. (2023) highlighted that introducing biodegradable polymeric coagulants improved floc stability and settling kinetics, achieving up to 90% total suspended solids (TSS) removal while also reducing chemical residue generation compared to traditional alum-based coagulation. Thus, while sedimentation, when used in isolation, is limited in efficiency and space utilization, integrating it with modern nanotechnological and membrane-based systems has revitalized its role as a cost-effective, sustainable primary treatment strategy, particularly in large-scale and decentralized wastewater management frameworks (Ashraf et al., 2024).

3.2 Flocculation

Flocculation uses flocculants to agglomerate dispersed particles into larger, settleable flocs. It improves the removal of colloidal and suspended particles, enhances solid–liquid separation, and can be combined with other methods (Wang et al., 2025a, b, c). However, it raises operational costs due to chemical additions. Another risk is inefficient floc formation from poor mixing or overdosing, leading to sludge that requires further treatment. Coagulation destabilizes colloidal and suspended particles by neutralizing their surface charges. It effectively removes turbidity, suspended solids, and colloidal particles. Coagulation does not effectively remove dissolved contaminants (Zhao et al., 2021).

3.3 Chemical Precipitation

Chemical precipitation is a widely employed physicochemical method for the removal of dissolved inorganic contaminants, particularly heavy metals, phosphates, and fluorides, through the formation of insoluble precipitates (Wang et al., 2021). While this process is operationally simple, effective, and can achieve high removal efficiencies (up to 95% for metal ions such as Cu^{2+} , Pb^{2+} , and Cd^{2+}), it presents considerable challenges related to sludge generation and post-treatment disposal (Pohl, 2020). Recent assessments indicate that the chemical sludge produced during precipitation processes often contains concentrated levels of heavy metals and residual reagents, posing a risk of secondary pollution if not properly stabilized or managed. According to Wang et al. (2023), the leaching potential of metal-laden sludge can lead to groundwater contamination and soil toxicity, particularly under uncontrolled land-disposal conditions. Moreover, commonly used chemical reagents (e.g., lime, alum, and ferric salts) increase salinity and cause pH fluctuations in effluents, complicating downstream biological treatment (Samer et al., 2015).

To mitigate these issues, recent studies have explored sludge valorization and recovery approaches, including converting metal-rich residues into value-added adsorbents or catalysts, thereby promoting a circular resource recovery model. Integrating precipitation with biogenic nanoparticle systems or membrane-assisted hybrid processes has also been shown to reduce sludge volume by nearly 40%, improve removal efficiency, and minimize the risk of secondary environmental contamination (Wang et al., 2023). Thus, while chemical precipitation remains a reliable treatment method for specific pollutants, sustainable sludge management and minimization strategies with biogenic nanoparticles are critical for reducing its environmental footprint and ensuring long-term viability in modern wastewater treatment systems (Sanni et al., 2024).

3.4 Trickling Filters

Trickling filters are effective for treating areas with high organic loading feeds and consume low energy and have simple operational and maintenance

requirements; hence, can easily integrate with other processes (Abyar et al., 2023). However, suspended solids and inorganic wastes pose challenges for these systems, potentially leading to clogging or short-circuiting of the filter bed. This is due to limited installation space and improper operational practices, which can result in odour generation (Rezai et al., 2021). Notably, the conventional wastewater treatment process has significant limitations: sedimentation struggles with fine particles, chemical precipitation generates excessive sludge, flocculation requires meticulous chemical management, and trickling filters are prone to clogging (Obi et al., 2024). These methods often fail to adequately address emerging contaminants. Conversely, adding biogenic nanoparticles offers a superior alternative, demonstrating high efficiency in contaminant removal and potentially overcoming the drawbacks of traditional techniques while reducing secondary pollution and energy consumption (Rath et al., 2025a, b).

4 New Innovative Strategies for Wastewater Remediation: Biogenic Nanoparticles for Sustainable Wastewater Treatment and their Mechanistic Synthesis Pathways

Unlike conventional chemical synthesis, biogenic approaches rely on the intrinsic biochemical machinery of plants, bacteria, fungi, or algae to reduce metal ions into nanoscale particles under mild, non-toxic conditions (Sheerswal et al., 2025). BNPs have emerged as a transformative class of materials precisely because their biomolecule-rich surfaces, redox-active cores, and structural tunability enable mechanisms inaccessible via conventional treatment systems (Tl et al., 2025). BNPs synthesized from plant extracts, bacteria, fungi, and algae show enhanced electron-transfer capabilities, abundant surface functional groups, and catalytic activity, enabling pollutant transformation in diverse wastewater systems (Rath et al., 2025a, b). For example, biogenic metal oxides (e.g., Fe_3O_4 , ZnO) demonstrate high adsorption capacities for Cr(VI) , Pb(II) , and As(III) in industrial effluents; biogenic TiO_2 and Ag nanoparticles exhibit strong photocatalytic and antimicrobial activity in pharmaceutical and hospital wastewater; and carbonaceous BNPs derived from microbial and algal biomass efficiently remove textile dyes and

polyaromatic hydrocarbons through π - π interactions and surface oxidation pathways (Gouda et al., 2025; Mabudi et al., 2025).

In plant-mediated synthesis, phytochemicals such as flavonoids, terpenoids, alkaloids, and phenolic acids act as both reducing and capping agents, facilitating the conversion of metal precursors into stable nanoparticles with controlled morphology and crystallinity (Shahzadi et al., 2025). Bacterial synthesis involves enzymatic reduction through nitrate reductases, hydrogenases, and metallothioneins, often yielding nanoparticles with uniform dispersion and high biocompatibility (Xie et al., 2025). Meanwhile, fungal-mediated pathways exploit extracellular secretion of enzymes and polysaccharides, yielding nanoparticles with superior surface functionalization and colloidal stability (Rath et al., 2025a, b). According to Wang et al., (2025a, b, c), the synergy between metabolic precursors and secreted biomolecules in these biological systems provides precise control over nucleation and growth kinetics, enhancing the catalytic and adsorptive properties of the resulting nanoparticles. Across the reviewed studies, BNPs consistently demonstrate advantages stemming from their biological capping agents, enhanced reactivity, and lower anticipated ecotoxicity compared with chemically synthesized nanomaterials (Mughal et al., 2021).

4.1 Nanoadsorbents

Nanomaterials are effective adsorbents and suit all classes of anionic and cationic pollutants. Nanoadsorbents possess a large surface area, variable pore size and surface chemistry, easy delivery, low-temperature modification, and short intraparticle diffusion distance. Like traditional adsorbents, biogenic nanoparticles adsorb contaminants via physicochemical mechanisms (Naser et al., 2021). Adsorption is the process by which materials (adsorbates) build up on the surface-active sites of an adsorbent through chemical or physical bonding. Various adsorbents can efficiently remove organic and inorganic pollutants without producing secondary contamination. The amount of adsorbate that a material can hold before reaching equilibrium is measured by its adsorption capacity (Mofokeng et al., 2024).

Biological macromolecules in the extract, whether from green synthesis or chemical reduction, reduce

and cap zero-valent metallic nanoparticles, which are then used as nano-adsorbents (Bakhtiari et al., 2024). Nanoparticles employed as adsorbents for heavy metal removal should be non-toxic, have a high adsorption capacity, be able to adsorb pollutants at low concentrations, be easily removed from the adsorbent surface, and be recyclable multiple times (Mancardi et al., 2023). The structure and morphology of nanoadsorbent materials also significantly affect their characteristics and efficiency (Malhotra et al., 2023). This adsorbent has numerous reactive sites, a large surface area, and the catalytic properties necessary for water remediation. Beyond morphology, surface chemistry influences how nanoadsorbents interact with harmful pollutants, determining mechanisms such as electrostatic attraction, hydrogen bonding, and specific chemical reactions. Surface functional groups and charge distribution in nano-adsorbents significantly affect their adsorption capacity, selectivity, and overall efficacy in removing harmful pollutants from wastewater, as shown in Fig. 2 (Nworie et al., 2024).

Adsorbents play a crucial role in treating biogenic wastewater through physical and chemical interactions. Physical adsorption is based on weak van der

Waals forces, which allow contaminants, such as organic molecules, to bind to the adsorbent surface. Chemical adsorption strengthens the interactions between pollutants and the nanoadsorbent, thereby enhancing the removal of heavy metals and certain organic compounds (Alijamali et al., 2021).

Ion exchange is a type of chemical adsorption in which ions are exchanged between the adsorbent and wastewater. It is used to remove pollutants such as dyes and heavy metals. Some adsorbents incorporate biological processes to removal contaminants through microbial metabolism and surface colonization, which is particularly beneficial for removing organic pollutants and pathogens (Velusamy et al., 2021). Effective biogenic wastewater management thus requires combining these mechanisms, each tailored to the specific pollutants present and aimed at achieving treatment outcomes (Dehghani et al., 2023).

Most nano-adsorbents used for wastewater remediation are carbon-based, including graphite, graphene oxide, and black carbon. Other adsorbents used for wastewater remediation include copper nanoparticles, iron oxides, gold, silver–gold, zinc oxide, silver, iron, and magnesium oxide (Sekar et al., 2023). Magnetic nanoparticles, especially iron oxide, have emerged as

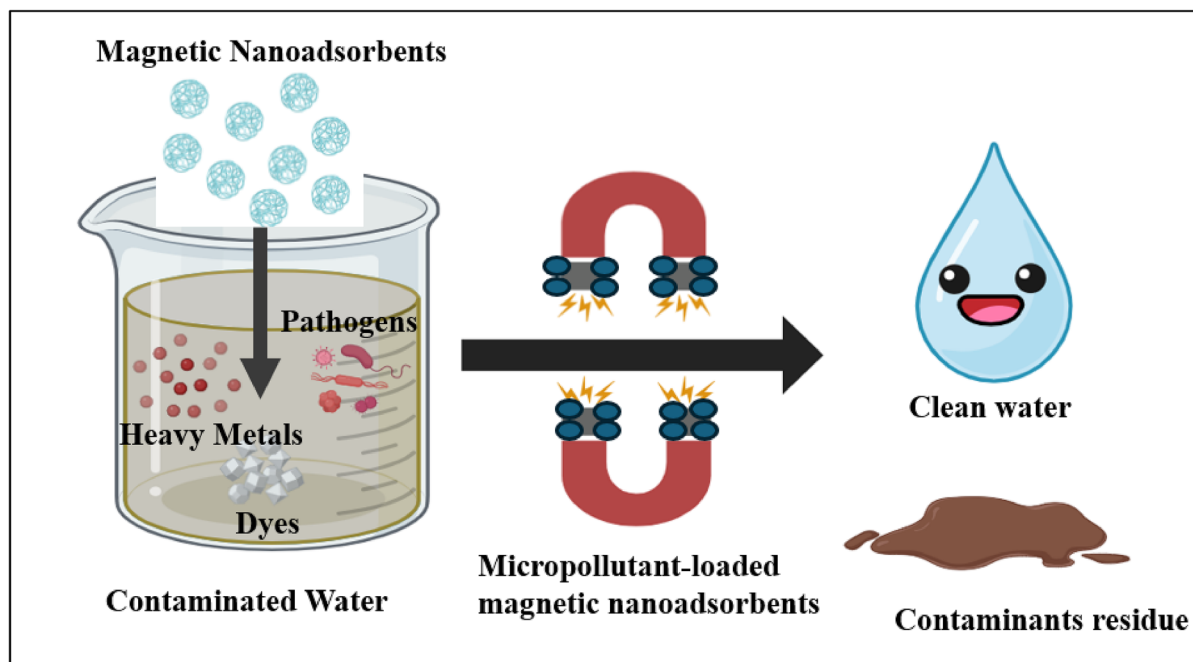


Fig. 2 Functions of nano-adsorbents in treating contaminated water

efficient nano-adsorbents for wastewater treatment due to their strong affinity for heavy metals and organic pollutants. Their key advantage is facile magnetic separation, enabling rapid recovery and reuse without generating secondary sludge (Fig. 2). Other examples include using gold nanoparticles as nano-adsorbents to detect and remediate heavy metals in water (Hua et al., 2021). Zeolite is another significant nano-adsorbent with a porous structure that can encapsulate various nanostructures, such as copper and silver ions. Zeolites can control metal concentrations while also acting as antibacterial agents. Moreover, zinc oxide nanoparticles are used to remove contaminants from wastewater, degrade dyes, and remove heavy metals, owing to their high oxidizing potential and large band gap (Dihom et al., 2022). Zinc oxide nanoparticles are efficient adsorbents for the removal of heavy metals, including mercury, arsenic, lead, cadmium, and chromium, as well as other hazardous metals.

Zirconium oxide nanoparticles synthesized by *Pseudomonas aeruginosa* are also used as nano-adsorbents and exhibit high adsorption capacity for tetracycline, thereby contributing to wastewater remediation (Zare et al., 2021). Copper nanoparticles are highly sensitive, exhibit adequate selectivity, have high adsorption efficiency for molecular oxygen, have a large surface area, and are low-toxicity, so these are used for the photocatalytic degradation of pollutants in wastewater. Nanoparticles of selenium, palladium, and platinum are present in an amorphous form and exhibit ligand-like adsorption processes, allowing them to successfully adsorb heavy metals, such as zinc, nickel, copper, mercury, and cadmium (Chandrakala et al., 2022).

Selenium (Se), palladium (Pd), and platinum (Pt) nanoparticles have attracted increasing attention for their exceptional catalytic, redox, and antimicrobial properties, making them promising candidates for advanced wastewater remediation (Barani et al., 2023). Biogenically synthesized Se nanoparticles exhibit strong antioxidative and photocatalytic activity, enabling efficient degradation of organic pollutants and detoxification of heavy metals. However, recent studies show that oxidative biotransformation of Se nanoparticles into selenite or selenate can induce cytotoxic and genotoxic effects in aquatic organisms at concentrations above threshold levels (Yu et al., 2025). These findings underline the need for controlled dosage and post-treatment monitoring

to prevent secondary environmental contamination. Similarly, lead (Pb) and platinum (Pt) nanoparticles exhibit excellent catalytic activity in hydrogenation and oxidation reactions, thereby enabling the degradation of recalcitrant organic compounds and persistent dyes. Despite their high functional efficiency, concerns remain regarding their long-term stability and potential ecotoxicity. Prolonged environmental exposure can cause surface oxidation, aggregation, and partial ion release, altering nanoparticle reactivity and reducing catalytic performance over time (Yu et al., 2025). Furthermore, chronic exposure to Pd and Pt nanoparticles has been linked to oxidative stress and inflammatory responses in aquatic biota, underscoring the importance of lifecycle assessment and safe disposal strategies (Leso et al., 2018).

To mitigate these challenges, recent approaches focus on surface modification and biomolecule capping to enhance the stability, dispersion, and biocompatibility of selenium, lead, and platinum nanoparticles (Javed et al., 2020). Incorporating natural polymeric stabilizers (e.g., chitosan, alginate) and proteinaceous capping agents derived from microbial or plant extracts has been shown to prolong nanoparticle stability for over six months under ambient conditions while reducing leaching and toxicity (Zare et al., 2021). Thus, future research should prioritize optimizing green synthesis and ecotoxicological evaluation to ensure the sustainable deployment of noble and metalloid biogenic nanoparticles in wastewater treatment systems.

The Langmuir isotherm model assumes monolayer adsorption on a homogeneous surface with a finite number of binding sites, representing systems in which adsorption occurs uniformly, without interactions between adsorbed molecules (Butyrskaya et al., 2024). Conversely, the Freundlich isotherm model accounts for multilayer adsorption on heterogeneous surfaces, indicating non-uniform energy distribution and varying adsorption affinities. The applicability of each model depends on the physicochemical nature of both the nano-adsorbent and the pollutant species (Isiuku et al., 2021).

Recent studies have demonstrated that biogenic metal oxide and carbon-based nano-adsorbents follow the Langmuir model for heavy metal removal (e.g., Cu^{2+} , Pb^{2+}), indicating monolayer binding, whereas organic dye adsorption often aligns with the Freundlich model due to complex surface interactions (Helmi et al., 2025; Panda et al., 2025). Despite

their high adsorption efficiency, key limitation lies in regeneration and reusability. Repeated desorption–adsorption cycles may cause surface fouling, agglomeration, and loss of active sites, reducing long-term performance (Edirisooriya et al., 2025). However, regeneration techniques such as mild acid/base washing, thermal activation, or photocatalytic regeneration have been shown to recover up to 80–90% of the original adsorption capacity after multiple cycles, making biogenic nano-adsorbents a sustainable alternative for continuous wastewater treatment applications (Jin et al., 2025).

4.2 Nanocatalysts

Nanocatalysis is an innovative approach that uses nanoparticles and plays a significant role in wastewater remediation, alongside adsorption, by converting contaminants into more innocuous forms (Mallikarjunaiah et al., 2020). It degrades persistent organic compounds by transferring electrons to contaminants, causing them to break down into benign molecules (Abbas et al., 2017). Biogenic nanoparticles have shown superior catalytic performance in eliminating

hazardous, refractory organic contaminants. Industrial processes that use pesticides, organic or inorganic solvents, pharmaceuticals, and hazardous metals generate most of these pollutants (Fig. 3).

In nanocatalysis, light interacting with metallic nanoparticles leads to high, broad-spectrum photocatalytic activity. Thus, nanophotocatalysts can aggregate in water and air, limiting their nanocatalytic activity. Their high size-to-volume ratio and shape-dependent characteristics enhance catalytic responsiveness and make them useful for wastewater remediation (Mohd Raub et al., 2024).

Copper oxide nanoparticles, zinc oxide nanoparticles, iron oxide nanoparticles, and titanium oxide nanoparticles are used for the photocatalytic degradation of various substances, including dyes, heavy metals, pesticides, chlorinated and non-chlorinated hydrocarbons, and aromatic compounds, into harmless byproducts (Rathod et al., 2024). Biogenic nanoparticles, such as cadmium sulfide nanoparticles synthesized from *Trichoderma harzianum*, act as photocatalysts for degrading methylene blue under UV light. Cadmium sulfide nanoparticles destroyed 38% of the methylene blue dye after 60 min. This

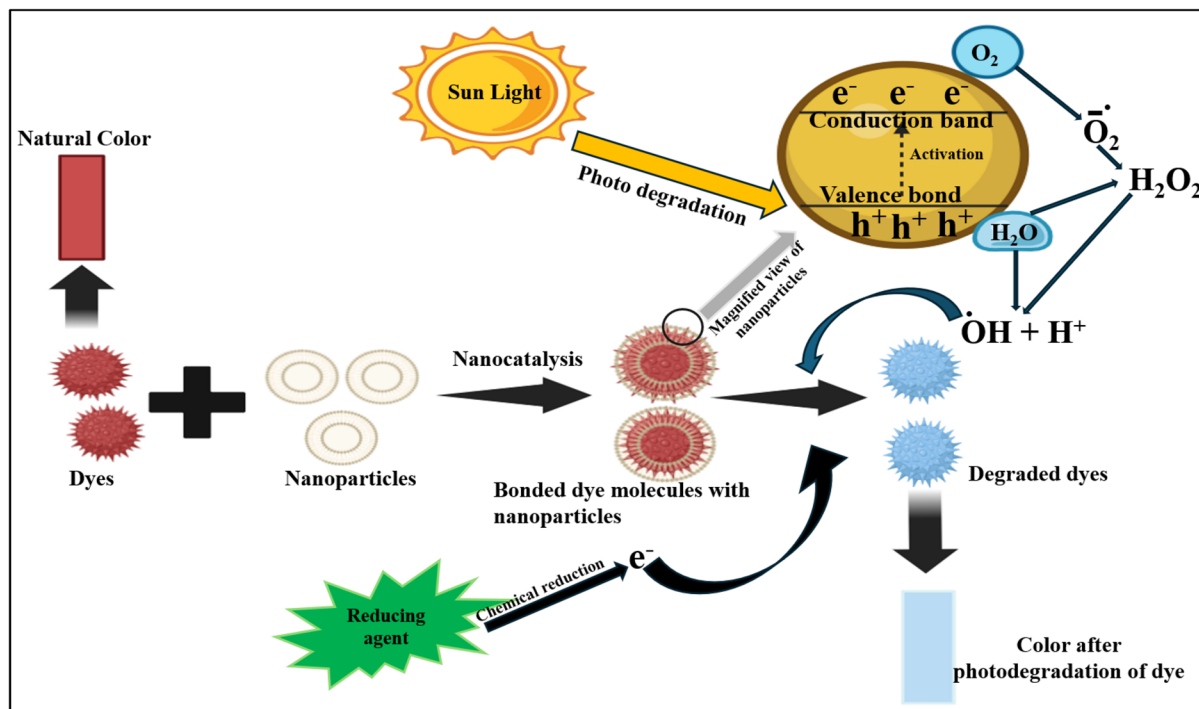


Fig. 3 Functions of nanocatalyst for the remediation of polluted water

increased activity was attributed to a high surface-to-volume ratio, which enhanced photocatalysis. Moreover, zinc oxide nanoparticles are nanophotocatalysts used for the removal of *E. coli* from wastewater (Rather et al., 2023).

Biogenically synthesized nanocatalysts, particularly those based on TiO_2 , ZnO , Fe_3O_4 , and CuO , exhibit superior photocatalytic and Fenton-like reactivity under mild conditions, enabling efficient mineralization of pollutants into less harmful end products such as CO_2 and H_2O (Kabir et al., 2025). However, beyond their catalytic efficiency, the environmental fate and post-treatment stability of nanocatalysts have emerged as critical factors influencing their sustainability. According to Jiang et al. (2025), prolonged exposure of catalytic nanoparticles in aquatic systems can lead to surface oxidation, partial dissolution, and ion leaching, potentially releasing reactive metal ions, such as Cu^{2+} , Fe^{2+} , or Zn^{2+} , into the environment. These released ions may cause secondary toxicity, disrupt microbial balance, and induce oxidative stress in aquatic organisms. Moreover, aggregation of nanocatalysts under natural water conditions can reduce their active surface area, thereby altering their catalytic efficiency and transport dynamics in sediments (Onotu et al., 2025).

Recent studies emphasize the importance of nanocatalyst recovery, immobilization, and lifecycle management to mitigate these environmental risks (Damian et al., 2025; Fahim et al., 2025). Strategies such as magnetic recovery, polymer encapsulation, and surface functionalization with biopolymers or green ligands have been shown to significantly reduce leaching and enhance reusability. Jiang et al. (2025) showed that immobilized nanocatalysts retained over 90% of their catalytic efficiency after five cycles while releasing negligible metal (<2%) into treated effluents. Therefore, while nanocatalysts remain a cornerstone of advanced oxidation and photocatalytic treatment technologies, future research must integrate fate assessment, ecotoxicological monitoring, and circular recovery mechanisms to ensure safe, sustainable application in real-world wastewater systems.

4.3 Nanomembranes

Nanomembranes use nanoparticles to improve toxin removal from wastewater by separating particles from the water. Due to their smaller pore size,

nanomembranes can filter out even the smallest particles, including viruses and bacteria (Shukla et al., 2024). Nanomembranes are typically 100 nm thick, selectively permeable, and highly uniform, offering efficiency, reduced processing time, and ease of handling, making them suitable for water remediation (Bhoj et al., 2021). Nanomembranes effectively remove dyes, heavy metals, and other contaminants from wastewater (Fig. 4). To enhance specific properties, researchers blend biogenic nanocomposite membranes with synthetic and biological materials. These adaptable membranes are used in environmental applications (pollutant removal and water purification) and in biomedicine (drug delivery, antimicrobial coatings, and wound healing). Their multilayer construction maintains filtration effectiveness while strongly rejecting unwanted materials. Due to their affordability and selectivity, nanomembranes, which fall between ultrafiltration and reverse osmosis, hold great promise for removing heavy metals (Chauke et al., 2025).

Various types of nanotubes, nanofibers, and nanoribbons function as membranes for wastewater remediation. These nanostructures have large surface areas and high reactivity, enabling effective adsorption and catalytic degradation of contaminants in wastewater (Das et al., 2024). Furthermore, their small size facilitates easier dispersion throughout the water matrix, increasing contact with pollutants. Additionally, their tunable features allow modification for specific contaminants, making them well-suited to various treatment processes, including filtration, adsorption, and photocatalysis (Khan et al., 2022).

The mechanism behind using a nanomembrane for wastewater remediation involves both physical and electrostatic interactions. Impurities larger than the membrane pore size are physically blocked, while charged particles are repelled or attracted depending on the membrane's surface charge (Tripathy et al., 2023). This selective filtration technique effectively removes impurities, yielding treated water that meets quality standards for reuse or environmental discharge. Nanomembranes consequently offer a long-term, efficient solution for treating biogenic wastewater, thereby conserving water and protecting the environment (Alshehri et al., 2025).

Advances in nanotechnology have made fabricating nanocomposite membranes increasingly important and efficient (Esfahani et al., 2019). As a result, researchers

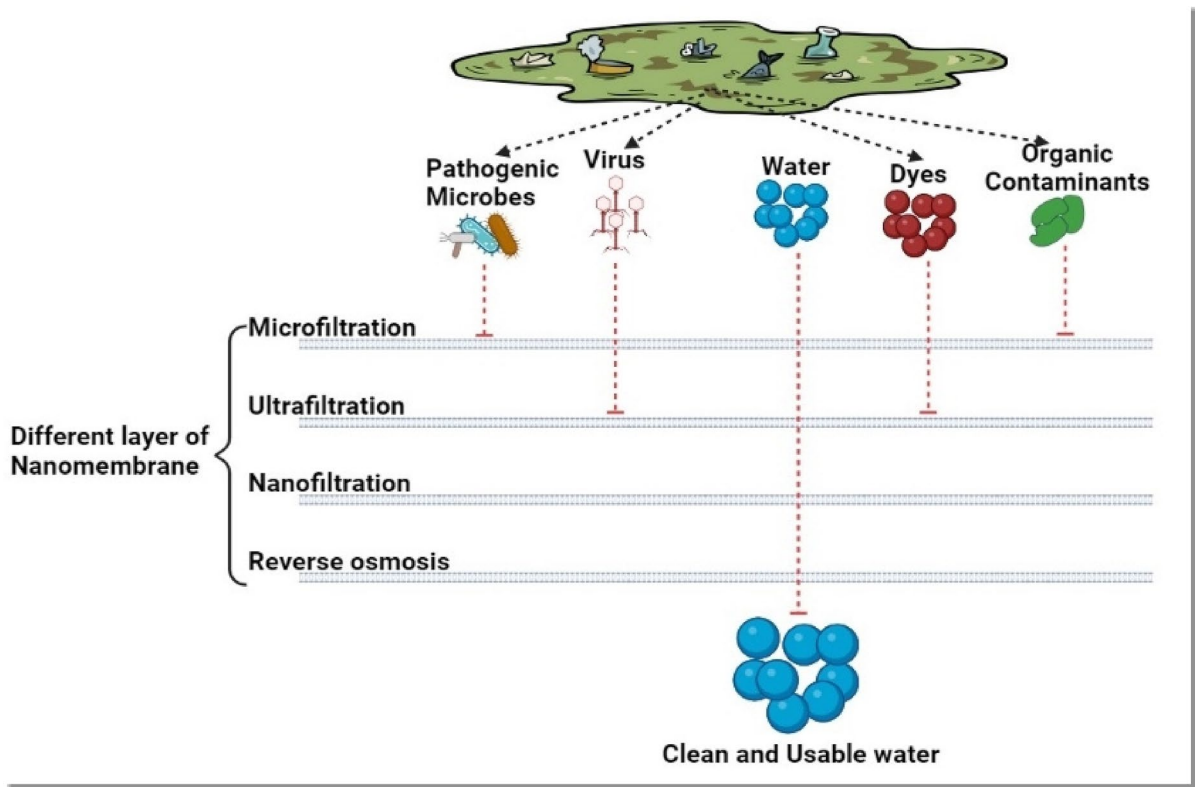


Fig. 4 Function of nanomembranes for filtration of contaminated water

have investigated a wide variety of nanoparticles for their effects on the engineering capabilities of polymeric nanocomposite membranes. In many cases, these nanoparticles have dramatically improved mechanical, thermal, and antifouling qualities. For specific applications, researchers often spread the composite layer into a polymer mold (Hsissou et al., 2021). Examples of nanomembranes include titanium oxide-doped nanofilms, which are used to degrade organic pollutants and in the removal of certain microorganisms. Moreover, zinc oxide nanocomposites synthesized from *Azadirachta indica* leaves exhibit antimicrobial activity (Parmar et al., 2019). Membranes are widely utilized in membrane bioreactor systems for wastewater treatment. Compared with traditional processes, membrane bioreactors combine activated sludge with membrane filtration, and generates significantly superior effluent quality, a smaller footprint, a higher organic loading rate, and less sludge.

Despite these advantages, membrane fouling remains a major operational bottleneck that critically

limits long-term performance and scalability (Zhang et al., 2025). Fouling results from the progressive deposition and adsorption of organic macromolecules, colloidal matter, inorganic precipitates, and microbial biofilms on the membrane surface or within its pores (Liu et al., 2025a, b). This phenomenon induces pore blockage, increases transmembrane pressure, and reduces permeate flux, ultimately reducing filtration efficiency and shortening the membrane's operational lifespan (Diez et al., 2020).

Recent investigations have identified fouling as the predominant cause of membrane performance deterioration, accounting for more than 60% of the total flux loss during continuous operation, particularly under high organic-load conditions (Chen et al., 2025). To address this limitation, researchers have developed various antifouling strategies using nanostructural and surface-engineering approaches. Functionalizing membrane surfaces with hydrophilic nanoparticles (such as TiO_2 , SiO_2 , and GO) enhances surface wettability, reduces foulant adhesion, and

facilitates smoother water transport pathways (Liang et al., 2025). Furthermore, incorporating antimicrobial nanoparticles, including silver (Ag) and copper oxide (CuO), has been shown to inhibit microbial colonization and biofilm formation, thereby prolonging membrane lifespan (Hosseinzadeh et al., 2025). An emerging frontier in membrane design involves integrating photocatalytic self-cleaning layers, in which semiconductor nanomaterials such as TiO₂ and ZnO generate reactive oxygen species (ROS) under ultraviolet or visible light irradiation, thereby promoting in situ degradation of organic and biological foulants (Xu et al., 2025).

Such nanocomposite modifications have been reported to enhance membrane durability by 30–50% while concurrently reducing the frequency of chemical cleaning and operational downtime (Rajendran et al., 2025). Moreover, hybrid treatment configurations that combine nanomembrane filtration with adsorption or photocatalytic oxidation exhibit synergistic antifouling effects, maintaining high flux stability and contaminant rejection across multiple operational cycles. As Jiang et al. (2025) highlight, incorporating nanomaterial coatings significantly mitigates leaching and oxidative deterioration, extending functional lifetimes and reducing environmental impact. Overall, continued advances in nanomaterial-assisted antifouling design and self-regenerative surface engineering are transforming nanomembranes into durable, low-maintenance, energy-efficient systems, paving the way for large-scale implementation in sustainable wastewater treatment infrastructure (Shahzad et al., 2021).

Nanofiltration is a very crucial aspect of nanomembrane systems, and involves a pressure-based separation technique with apparent pore sizes in the nanoscale range (0.5–2 nm) and a molecular weight of 200–1000 Da. For rejecting ionic or molecular species, nanofiltration lies between reverse osmosis and ultrafiltration. Nanofiltration membranes exhibit higher hydrodynamic resistance and are more compact in design (Rath et al., 2025a, b). Nanofiltration membrane preparation involves various techniques, including surface coating, surface graft modification, L-S phase change, interfacial polymerization, and layer-by-layer self-assembly (Azizi et al., 2022). Thin film nanocomposite membranes and electrospun nanofibrous nanofiltration membranes are a few types of nanofiltration membranes. Nanofiltration

membranes have become increasingly important because of improvements in lifespan, permeate quality, and antifouling properties (Anand et al., 2018).

Polymeric thin-film nanocomposite nanofilters have been widely used as membrane materials in wastewater treatment; however, concerns remain about their fouling tendencies and long-term stability (Karami et al., 2020). Nanofiltration membranes have also played a crucial role in removing organic pollutants. Chlorine disinfection is crucial for eliminating microbial growth observed in nanofiltration distribution systems. Nanofiltration membranes, characterized by high removal of organic materials and low retention of inorganic materials, can produce high-quality water while reducing microbial growth (Allaoui et al., 2022).

5 Mechanistic Insights and BNPs Advantages

Modern wastewater consists of a complex mixture of heavy metals, dyes, organic pollutants, pharmaceuticals, and pathogenic microorganisms (Fig. 5). While nanomaterials have gained prominence as versatile tools for contaminant removal, the synthesis reveals that biogenic nanoparticles generated through plant, microbial, fungal, or algal pathways offer distinct mechanistic advantages over conventional nanomaterials and treatment methods (Kapoor et al., 2021).

5.1 Heavy Metal Remediation: Mechanistic Insights and BNPs Advantages

Heavy metals are non-biodegradable, toxic, and carcinogenic. Lead, zinc, mercury, nickel, cadmium, copper, chromium, and arsenic are the most commonly occurring heavy metals (Balali et al., 2021). Disposing of these heavy metals into water bodies significantly contributes to water pollution and poses a serious threat to both human health and the environment. However, with the expansion of industry and human activities, including mining, plating, electroplating, battery manufacturing, and pesticide production, heavy metal concentrations in wastewater have been increasing (Mishra et al., 2019). Even in small amounts, these heavy metals can adversely affect the environment and human health. Therefore, remediation of heavy metals in wastewater is crucial for environmental protection. Table 1 presents the

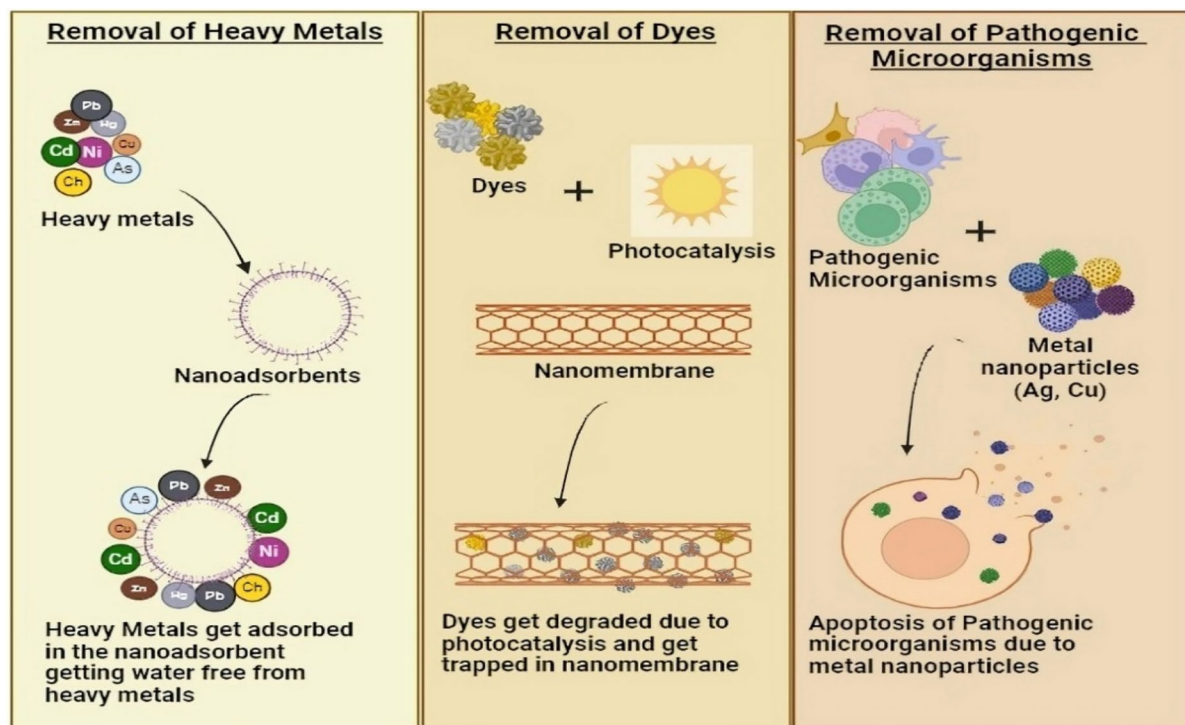


Fig. 5 Applications of biogenic nanoparticles in wastewater treatments

use of biogenic materials for heavy metal removal. These metals pose significant risks to ecological and human health even at trace concentrations, prompting international regulatory agencies to establish strict permissible limits. According to the World Health Organization (WHO, 2022) and the United States Environmental Protection Agency (USEPA,

2023), drinking water standards set the maximum permissible limit for cadmium at 0.003 mg/L, reflecting its high nephrotoxicity and potential to cause skeletal deformities. Hexavalent chromium (Cr^{6+}), known for its carcinogenicity and mutagenic potential, is regulated at 0.05 mg/L (WHO) and 0.1 mg/L (USEPA total chromium), and Australian Freshwater

Table 1 Different biogenic nanoparticles in pollutant removal

Sl. no	Nanoparticles	Synthesized from	Pollutants Removal	References
1	Iron	<i>Eucalyptus globules</i>	Phosphate	Gan et al., 2018
2	Iron	<i>Clonorchis sinensis</i>	Arsenite, Arsenate	Lunge et al., 2014
3	Copper	<i>Hibiscus sabdariffa</i>	Nitrate	Paixão et al., 2018
4	Copper	Citrus peel extract	Photocatalytic degradation of methyl red dye	Sinha & Ahmaruzzaman, 2015
5	Cadmium sulfide	<i>Pseudomonas aeruginosa</i>	Cadmium	Raj et al., 2016
6	Selenium	<i>Citrobacter freundii</i>	Mercury	Wang et al., 2018
7	Manganese	<i>Pseudomonas putida</i>	Lead, Cadmium	Zhou et al., 2015
8	Magnesium oxide	<i>Acacia</i> spp.	Nickel, Lead, Cadmium, Zinc, Cobalt, Copper	Srivastava et al., 2019
9	Gold	Plant leaf extract	Methylene blue, Methyl orange, Bromophenol blue	Choudhary et al., 2017
10	Silver	Petals extract	Degradation of Putnam sky blue 39 dye	Suárez-Cerda et al., 2015

guidelines ANZCECC and ARMCANZ (2000) are 0.001 mg/L in potable water (Nugegoda et al., 2013). Arsenic, a Group 1 carcinogen linked to dermal, cardiovascular, and neurological disorders, is set at an even more stringent limit of 0.01 mg/L by both WHO and USEPA to minimize chronic exposure risk (Jomova et al., 2011). Discharging untreated or partially treated industrial wastewater often results in metal concentrations that far exceed these thresholds, necessitating advanced remediation technologies to meet global regulatory standards. Thus, these regulatory benchmarks provide essential context for evaluating the efficacy of biogenic nanomaterials and other emerging treatment technologies for safe, sustainable heavy-metal remediation (Firinčă et al., 2025).

Despite extraction and storage challenges, biogenic ferrous sulfide nanoparticles (bio-FeS) show promise for removing heavy metals. By integrating self-assembled bio-FeS with bacterial cells, researchers created a self-regenerative bio-hybrid that effectively eliminates chromium (Cr(VI)). By forming a thick coating on *Shewanella oneidensis* MR-1 cells, bio-FeS creates an artificial transmembrane electron channel that enables continuous Cr(VI) reduction over several treatment cycles. This system offers a sustainable alternative for treating heavy-metal wastewater, efficiently handling both simulated and actual electroplating wastewater, with activity readily recovered through bio-FeS resynthesis (Fu et al., 2021).

Some techniques used for heavy metal removal include nanoadsorption and nanomembrane filtration. Commercial and polymeric nano-adsorbents can efficiently capture and remove heavy metal ions from wastewater (Qasem et al., 2021). Nanoadsorbent surfaces can absorb heavy metal ions. A recent study showed that iron nanoparticles derived from natural iron adsorb chromium, copper, zinc, and arsenic from wastewater (Thilakan et al., 2022). For instance, microbial- and plant-derived BNPs not only adsorb metal ions but also reduce high-valence toxic species, such as Cr(VI), to the less soluble and less toxic Cr(III). These mechanisms are particularly effective in electroplating, tannery, and mining wastewater systems, where removal efficiencies of 70–95% have been consistently reported. Such activity directly addresses limitations of conventional precipitation and flocculation, which often fail at trace concentrations or produce excessive sludge (Wang et al., 2025a, b, c).

One of the most common heavy metal pollutants is hexavalent chromium (Cr⁶⁺), which adversely affects aquatic environments (He et al., 2020). Eliminating Cr(VI) from wastewater before release is crucial, as excessive amounts can harm the respiratory tract, induce various malignancies, and be fatal. Bacterial species such as *Shewanella* and *Geobacter* can synthesize biogenic palladium nanoparticles that efficiently remove chromium from wastewater. High concentrations of arsenic, iron, and manganese are present in both groundwater and surface water, which can impact human health and aquatic ecology (Tong et al., 2021). *Pseudomonas QJX-1* extracted from the manganese mines is capable of oxidizing manganese (II) and is utilized in the production of biogenic iron–manganese oxides. These oxides can both oxidize and adsorb arsenic (Ye et al., 2023).

Another extremely hazardous heavy metal naturally present in the environment is cadmium. Contamination of water with cadmium causes diseases, such as itai-itai, and too much consumption of Cd²⁺ can cause other health-related risks, which include high blood pressure, bone marrow cancer, renal damage, and gastrointestinal problems. Biogenic nanoparticles have been applied extensively to remove cadmium from water. EPS, or Extracellular polymeric substances of the marine bacteria *P. aeruginosa JP-11*, were used to synthesize cadmium nanoparticles, which were then utilized for preconcentrating Cadmium from simulated wastewater (Ghosh et al., 2022).

To remove uranium from wastewater, researchers developed regenerable biohybrids comprising biogenic mackinawite nanoparticles (BX-FeS) stabilized with xanthan gum (XG). To yield smaller, more stable, and more active spherical FeS nanoparticles, XG functions as a carrier that promotes Fe²⁺ attachment. After two years of storage, BX-FeS remained effective. It demonstrated a uranium removal efficiency of 97.9% across a wide pH range and in the presence of competing ions, representing a substantial improvement over the 67.1% achieved by biogenic B-FeS. At a lower cost than substitutes such as silica gel and zerovalent iron, the material demonstrated outstanding adsorption capacity (658.0 mg/g) for uranium in effluent. Crucially, BX-FeS prevented secondary pollution from Fe³⁺ and SO₄²⁻ while sustaining high activity during three regeneration cycles driven by *Desulfovibrio desulfuricans* (He et al., 2024).

5.2 Organic Pollutants and Pharmaceuticals:

Catalysis-Driven BNPs Transformation Pathways

Rapid urbanization has significantly increased organic pollutant concentrations in water. Organic pollutants encompass a range of substances, including xenobiotics such as insecticides, synthetic dyes, pesticides, aromatic hydrocarbons, pharmaceuticals, halogenated hydrocarbons, and phenols (Akhtar et al., 2021) (Fig. 6). Human health and the environment will be negatively affected by trace amounts of organic pollutants found in soil, water, air, and even food (Madhav et al., 2020). To eliminate organic pollutants, nano-enabled technologies use novel products such as nanoadsorbents, nanocatalysts, and nanofilms (Fig. 6). Organic environmental pollutants can also be monitored using nanobiosensors in which the functional element includes a transducer element that signals the downstream event and a recognition element that binds to the target component (Ekrami et al., 2021).

The degradation of persistent micropollutants is facilitated by biogenic palladium nanoparticles (bio-Pd NPs). An electrochemical cell that generated H₂ as the electron donor enabled the regulated synthesis of bio-Pd NPs of various sizes (Law et al., 2022). Superior catalytic activity was demonstrated by larger NPs (39.0 nm) generated at lower H₂ flow rates, which removed 92.1% of the methyl orange, whereas smaller NPs (23.2 nm) removed only 44.3%. The 39.0 nm bio-Pd NPs successfully removed eight micropollutants at environmentally relevant concentrations (ng/L to µg/L) when applied to secondary-treated municipal wastewater. The removal efficiencies ranged from 69.5% (ibuprofen) to 97.2% (lorazepam). More than 90% of fluorinated antibiotics were eliminated (Law et al., 2023).

Unlike the previously discussed inorganic pollutants, these compounds exhibit complex molecular structures, high chemical stability, and resistance to conventional biological degradation, which contribute to their long-term persistence in aquatic ecosystems (Thakur et al., 2025). Many organic pollutants, including polycyclic aromatic hydrocarbons (PAHs), synthetic dyes, endocrine-disrupting chemicals (EDCs), and pharmaceutical active compounds (PhACs), are highly hydrophobic, facilitating adsorption onto sediments and biomagnification across trophic levels (Zou et al., 2021). Molecular polarity, functional group chemistry, and electron-withdrawing substituents govern their environmental behavior by influencing reactivity and interactions with

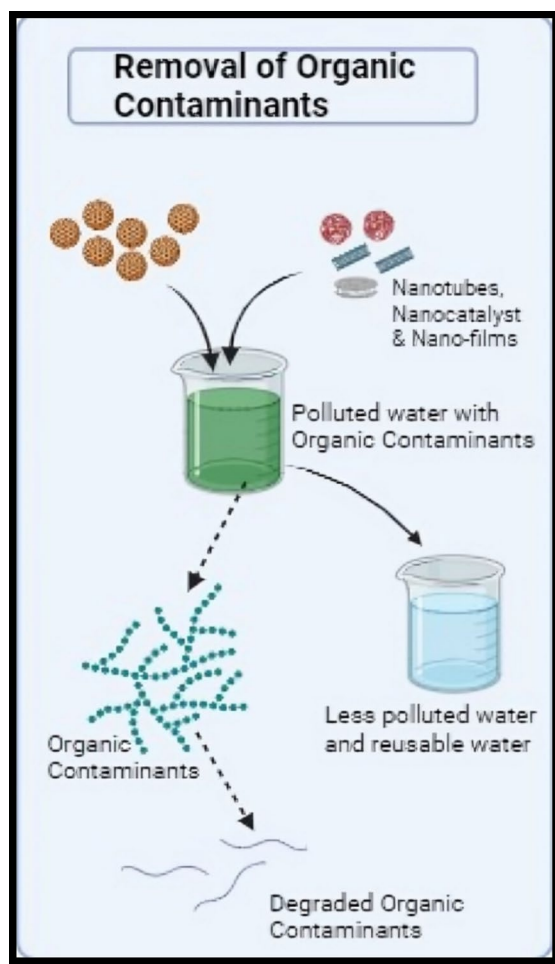


Fig. 6 Removal of organic contaminants using nanoparticles

advanced nanomaterials used in wastewater treatment. BNPs play a synergistic role not only in promoting degradation but also in accelerating mineralization and reducing toxic intermediate formation, a limitation commonly observed in classical photocatalysis (Hamza et al., 2024). In pharmaceutical wastewater treatment, BNPs have achieved 80–95% degradation within 60–120 min, surpassing conventional oxidation processes that often require chemical additives or high-energy inputs (Qin et al., 2019).

5.3 Dye Remediation: BNP-Enabled Adsorption–Catalysis Synergy

Dyes are a major concern because of their adverse effects on the environment and human health. When

wastewater from dyed materials is released into the environment, it can disrupt natural ecosystems (Karri et al., 2021). Toxic dyes and effluents from the textile and dyeing industries disrupt aquatic plant photosynthesis by inhibiting photosynthetic pigments, reducing light penetration, and increasing chemical and biological oxygen demands in water bodies. If dyes enter the food chain, can have allergic, mutagenic, or carcinogenic effects on living organisms. Thus, a major global concern is efficiently removing harmful dyes from wastewater (Rathi et al., 2021).

Various physicochemical methods, including ozonation, membrane filtration, adsorption, ultrafiltration, and advanced oxidation processes via photocatalysis or chemical reduction, can treat textile effluents and address, this issue. Because of their exceptional stability, these methods are ineffective for degrading dyes (Katheresan et al., 2018). Because metallic nanoparticles have unique biological and catalytic properties, have recently been shown to be effective tools for wastewater treatment. Thus, catalytic degradation is considered one of the most effective techniques for dye degradation. By photocatalysis, harmful compounds can be converted into less hazardous or non-toxic ones. Reports have described the use of several metal nanoparticles, including silver, gold, platinum, palladium, and iron, as catalysts for degrading various harmful dyes (Harikumar et al., 2022).

Removal of methylene blue via biological processes is challenging due to its toxicity. The algal species *Chlorella pyrenoidosa* was used to biogenically synthesize silver nanoparticles (Ag NPs), which exhibited nanophotocatalytic activity in remediating methylene blue. Biogenic silver nanoparticles outperformed commercially synthesized silver nanoparticles in photocatalytic degradation of methylene blue (Ifijen et al., 2022). Remarkably, *E. harbicola* biomass produced selenium nanoparticles that strongly absorbed methylene blue from wastewater via photocatalytic degradation. Biogenic zinc oxide nanoparticles derived from *Hibiscus rosa-sinensis* leaf extract can adsorb Congo red from wastewater, thereby playing a crucial role in wastewater remediation (El-Behery et al., 2023).

Ag NPs synthesized via green methods using *Linum usitatissimum* extract demonstrate multiple applications and achieved 67.13% photodegradation of methylene blue, demonstrating effective photocatalytic performance for wastewater

treatment. Additionally, these biogenic Ag NPs serve as enzyme-free electrochemical sensors for hydrogen peroxide detection, exhibit potential as antibacterial and anticancer agents, and thus represent versatile biological agents for environmental and medical applications (Karimi et al., 2023). Ag NPs were synthesized from *T. foenum-graecum* (fenugreek) seed extract for the degradation of dyes such as methyl orange, methylene blue, and eosin Y. To effectively remove dyes such as 4-nitrophenol, methylene blue, methyl orange, phenol red, and direct blue 24, which are harmful to human health and the environment, silver nanoparticles derived from *S. acuminata* fruit extract can be used (Patel et al., 2021). In a fermentative environment, *Klebsiella oxytoca* GS-4-08 produced palladium nanoparticles that reduced azo dye concentrations in wastewater (Ma et al., 2022). To extract methyl orange (MO) colour from water, researchers created bimetallic core-shell Pd/Fe nanoparticles on aerobic microbial granules (Bio-Pd/Fe). The granules produced Bio-H₂, which catalyzed Bio-Pd/Fe synthesis. Bio-Pd/Fe achieved 99.33% MO removal under optimal circumstances identified by response surface methods, which was much better than Bio-Pd (68.91%), Bio-Fe (76.8%), and other controls. The analysis demonstrated the efficacy of Bio-Pd/Fe for dye removal by confirming MO degradation into less hazardous compounds, a finding corroborated by bioassays using freshwater algae and *Allium cepa* (Kubendiran et al., 2021).

Photocatalysts such as TiO₂, ZnO, and Fe₂O₃ nanoparticles generate reactive oxygen species under light irradiation, initiating partial oxidation of complex dye molecules and breaking them down into smaller, more biodegradable intermediates (Senthil et al., 2024). These transformed intermediates are then mineralized by dye-degrading microbial consortia, including species of *Pseudomonas*, *Bacillus*, and *Aspergillus*, which possess enzymatic pathways for azoreductase-, peroxidase-, and laccase-mediated degradation (Rajendran et al., 2025). Hybrid systems, such as TiO₂-assisted photocatalysis coupled with *Pseudomonas putida* or ZnO nanoparticle pretreatment followed by fungal biodegradation, have achieved 95–98% decolorization and > 80% reduction in total organic carbon (TOC), out-performing standalone photocatalytic or biological processes (O'Neill et al., 2023). Moreover, immobilizing

photocatalysts on biocompatible substrates enhances microbial viability and reduces nanoparticle leaching, thereby improving operational stability and ecological safety of these integrated systems. Such hybrid nanotech–biological approaches offer a promising, sustainable strategy for treating dye-laden wastewater, especially when traditional methods are inefficient (Fahim et al., 2025).

5.4 Microbial Pathogen Reduction: BNPs Antimicrobial Mechanisms in Wastewater Systems

Using biogenic metal nanoparticles to removal pathogenic microorganisms is a promising and sustainable approach. Microorganisms, including bacteria, actinomycetes, and fungi, can absorb precursor metal ions and produce corresponding nanoparticles as a detoxification mechanism (Rocha et al., 2024). This method is cost-effective, environmentally benign, and does not require capping or stabilizing agents, making it an attractive option for pathogen removal (Kapoor et al., 2021).

The antimicrobial properties of metal nanoparticles, such as silver nanoparticles, are well investigated. Metal nanoparticles disrupt pathogen cell walls, increase permeability, and show strong antimicrobial activity against a wide range of Gram-negative and Gram-positive bacteria (Prasad et al., 2017). Furthermore, combining metallic nanoparticles with antibiotics has shown synergistic effects against bacteria, including multidrug-resistant strains, and may reduce the emergence of antimicrobial resistance (Singh et al., 2020). In wastewater treatment, biogenic metal nanoparticles, such as magnetite nanoparticles produced by microorganisms, have been explored for the removal of heavy metals and bacterial pathogens. These nanoparticles have demonstrated high efficiency in removing toxic metal ions and pathogenic bacteria, making them promising candidates for water treatment applications (Ojha et al., 2020).

Exposure to nanoparticles can induce oxidative stress, membrane disruption, and metabolic inhibition, thereby contributing to pathogen inactivation; however, these same stress conditions may also promote horizontal gene transfer (HGT) among surviving microbial populations (Ngoepe et al., 2025). Stress-induced secretion of extracellular polymeric substances (EPS), together with increased cell membrane permeability, can facilitate the exchange of mobile genetic elements,

including plasmids and integrons that carry antimicrobial resistance (AMR) genes (Balta et al., 2025). Furthermore, prolonged or sublethal exposure to metal- and metalloid-based biogenic nanoparticles may exert selective pressure on environmental microbes, enriching resistant strains and accelerating the dissemination of antimicrobial resistance genes—key determinants in wastewater ecosystems (Balta et al., 2025). Recent studies report that nanoparticle-induced reactive oxygen species can activate bacterial SOS responses, increase mutation rates, and enhance the likelihood of resistance gene propagation (Shao et al., 2025). Therefore, while biogenic nanoparticles offer promising antimicrobial capabilities, their deployment must be accompanied by comprehensive AMR risk assessment, monitoring of gene transfer events, and the development of controlled-exposure strategies to mitigate unintended ecological and public-health consequences (Murugan et al., 2025).

In summary, using biogenic metal nanoparticles to removal pathogenic microorganisms offers a sustainable and effective approach (Rath et al., 2025a, b). Microbial synthesis of these nanoparticles offers advantages such as environmental benignity and high biocompatibility. Furthermore, the antimicrobial properties of metal nanoparticles suggest their potential for developing effective wastewater treatment strategies. Further research and development in this field could lead to the widespread application of biogenic metal nanoparticles in water treatment and public health initiatives (Wahab et al., 2023).

6 Metal Oxide Nanoparticles in Wastewater Remediation

Researchers use pure metal precursors to synthesize metal oxide nanoparticles (Jeevanandam et al., 2018). Due to their dynamic properties, these metal oxide nanoparticles are extremely useful for various environmental applications, including the removal of hazardous azo dyes, heavy metals, pesticides, and colorants from wastewater and have promising applications in environmental contexts (Senthil et al., 2024).

Applications of metal and metal oxide nanoparticles span various industries, including molecular sensing, environmental remediation, medicine, and catalysis. Metal nanoparticles include manganese oxides, nanosized iron oxides, titanium oxides, cerium

oxides, zinc oxide, magnesium oxides, aluminum oxides, and zirconium oxides (Jain et al., 2022). Metal nanoparticles exhibit well-known localized surface plasmon resonance, which confers distinctive optoelectronic properties and makes them suitable for various research fields (Marinescu et al., 2020). Among the many properties of nanosized metal oxides exhibit high removal capacity and selectivity for heavy metals. These application-focused examples highlight the significant potential of metal oxide nanoparticles to address pollutant-specific challenges in complex wastewater matrices and reinforce their role as high-efficiency, scalable solutions for modern wastewater remediation (Swathika et al., 2025).

Incorporating metallic nanoparticles into a wide range of industrial products exposes them to various environmental media, including air, water, oil, and renewable energy sources. Therefore, biogenic synthesis of these metal oxide nanoparticles is crucial because it has no negative environmental or human health impacts (Gebre et al., 2019). For example, *Ocimum sanctum* leaf extract was used to biogenically synthesize nickel nanoparticles and remove industrial dyes, including sulfate, nitrate, anionic pollutants, Orange II azo, crystal violet, and eosin Y (Fatima et al., 2025). Researchers studied zinc oxide nanoparticles synthesized from plant extracts of *Hibiscus subdarifa*, *Azadirachta indica*, and *Aloe vera* for their potential bactericidal and antifungal activities and as photocatalysts for degrading organic dyes (Basnet et al., 2018).

Zinc oxides are excellent photocatalysts because of their high chemical stability and strong photocatalytic activity in removing water contaminants. Zinc oxide nanostructures come in various forms, including nanosheets, nanorods, nanowires, and complex hybrid structures. Zinc oxides are among the most promising catalysts for remediating contaminated water because of their unique qualities, including availability, low cost, and non-toxicity (Mohd Adnan et al., 2016). In a recent study, zinc oxide nanoparticles were synthesized from *Acalypha indica* leaf extract, and their potential antibacterial properties, required for wastewater remediation, were examined (Karthik et al., 2017). Similarly, copper oxide nanoparticles possess a range of advantageous chemical and physical properties, including high-temperature superconductivity, solar energy efficiency, relative stability, low cost, and antibacterial activity. When methylene blue dye was subjected

to photocatalytic degradation in an aqueous *Aloe vera* solution at room temperature, the resulting copper oxide nanoparticles exhibited an active bandgap in visible light, and the dye was effectively degraded (Jabeen et al., 2024). Synthesized copper oxide nanoparticles have demonstrated strong antibacterial activity against *Escherichia coli* and *Staphylococcus aureus* and are resistant to methicillin at minimal bactericidal concentrations in wastewater (Naseem et al., 2021).

7 Fate of Biogenic Nanoparticles after Purification

The environmental fate of BNPs following wastewater treatment is governed by their partitioning, transformation, and long-term mobility in sludge and effluent. Studies indicate that 60–75% of BNPs accumulate in sewage sludge, while 25–40% remain dispersed in the treated effluent, depending on particle size, capping agents, and wastewater chemistry (Liu et al., 2023). Although sulfidation of biogenic silver nanoparticles (Ag NPs), leading to their conversion into less soluble Ag₂S under anaerobic digestion within approximately 40 h, is a well-documented transformation pathway, recent research highlights that BNPs undergo multiple concurrent transformations, including oxidation, dissolution, aggregation, and formation of organometallic complexes with humic substances and natural organic matter (Hassan et al., 2023). These processes significantly influence nanoparticle bioavailability, toxicity, and persistence. Importantly, retaining BNPs in biosolids destined for land application has direct implications for soil quality and terrestrial ecosystems. In soils, BNPs may interact with minerals and organic ligands, altering their mobility, potentially releasing metal ions, or incorporating them into soil aggregates. Their long-term presence can influence microbial activity, nutrient cycling, and plant uptake pathways, particularly with repeated biosolids application (Denisse et al., 2021). Conversely, BNPs discharged in effluent can enter surface waters, where exposure to sunlight and dissolved organic matter can further transform them into new chemical species with distinct ecotoxicological profiles (Shiade et al., 2024).

Considerations must include sludge stabilization, leachate monitoring, biosolid application guidelines, effluent quality control, and the development of transformation-aware risk assessments. These expanded

insights underscore the need for comprehensive environmental management frameworks that address not only BNP performance in treatment systems but also their long-term ecological behavior after disposal (Khan et al., 2025a, b). The study found that particle size distribution changes substantially during treatment, with 70% of the initial 20–50 nm particles growing into larger aggregates (> 100 nm) (Mukhopadhyay et al., 2022). One crucial aspect to consider is the soil retention capacity of biosolids treated on land, as it directly affects their effectiveness in BNPs (Sánchez-Monedero et al., 2024). Research indicates that approximately 25% of BNPs gradually migrate to deeper soil horizons over 6–12 months, while the remaining 30–40% remain attached to soil particles in the top 15 cm. Soil pH and organic matter content significantly affect mobility. Acidic soils (pH < 6) exhibit 40% higher particle mobility than neutral soils (Pérez-Esteban et al., 2019).

Complex transformation pathways govern the fate of BNPs and dictate their reactivity, mobility, and long-term ecological impact (Dubey et al., 2024). While sulfidation, oxidation, and surface passivation are well-recognized transformation routes, recent studies have revealed that BNPs can also undergo organometallic and organocomplex formation in natural waters. This occurs through interactions with natural organic matter (NOM), humic and fulvic acids, and dissolved organic ligands, which possess abundant functional groups, such as carboxyl, hydroxyl, and phenolic moieties, that can coordinate to nanoparticle surfaces (Nanditha et al., 2025). These reactions can produce stable or semi-stable organometallic complexes by coordinating to nanoparticle surfaces via functional groups or active binding sites, thereby altering surface charge, colloidal stability, dissolution kinetics, and redox behavior (Kostova, 2023).

Such transformations significantly influence BNP bioavailability and organo-complexation and may either reduce toxicity by passivating reactive surface sites or enhance biological uptake by increasing nanoparticle dispersion and affinity for cellular membranes (Nanditha et al., 2025). Emerging evidence indicates that metal-based BNPs, particularly those composed of Ag, Cu, Fe, and Zn oxides, can bind strongly to NOM-derived ligands, altering aggregation behavior and slowing sedimentation rates, thereby increasing residence time in aquatic systems (Liao et al., 2017).

8 Barriers to Biogenic Nanomaterials in Water Treatment

Despite the potential benefits of biogenic nanoparticles in wastewater treatment, several significant challenges must be addressed before their widespread implementation (Sanni et al., 2024). These microorganisms' tendency to aggregate in complex wastewater mixtures poses a considerable challenge, as it reduces their surface area and hampers treatment efficiency (Sharma et al., 2023). Recent studies show that BNPs can lose up to 40–50% of their catalytic activity within the first 24 h of deployment (Lee et al., 2024). Furthermore, pH significantly affects BNP stability. Gautam et al. (2024) reported that silver BNPs are most effective at pH 6.5–7.5, with removal efficacy decreasing by 30% outside this range. Temperature fluctuations in wastewater treatment facilities can affect BNP performance. Typically ranging from 10 to 35 °C. For every 5 °C rise above 25 °C, pollutant removal efficiency decreases by approximately 15% (Delgado et al., 2019).

Scale-up challenges remain a significant hurdle. Laboratory studies show that 95% of organic contaminants and heavy metals can be effectively removed (Kumari et al., 2025). However, in pilot-scale implementations, efficiency has only reached 60–70%. Multiple ions and the varied characteristics of real wastewater systems are the primary factors contributing to this efficiency gap (Qasem et al., 2021). Given current production methods, which require an investment of \$200–\$300 per gram for high-quality BNPs, cost considerations are crucial. Despite efforts to recover and reuse particles, most systems achieve only a 40–50% success rate, making the process technically challenging (Maslin et al., 2021).

Beyond technical and environmental constraints, socioeconomic and community-level factors significantly influence the widespread adoption of biogenic nanoparticle-based wastewater treatment technologies (Choi et al., 2024). Public acceptance of nanotechnology remains mixed, largely because of concerns about perceived toxicity, nanoparticle release, and potential long-term ecological impacts. Studies have shown that communities with limited scientific literacy often express heightened apprehension toward novel nanomaterials, especially when their environmental fate is not clearly communicated. This underscores the need for transparent risk communication strategies, community

engagement, and regulatory assurance to build trust in BNP-based interventions (Huang et al., 2024).

Additionally, the economic feasibility and long-term scalability of BNP technologies pose notable challenges. Although biogenic synthesis is environmentally benign and cost-effective at the laboratory scale, transitioning to industrial-scale production requires consistent biomass availability, bioreactor optimization, and reliable quality control to ensure nanoparticle uniformity (Abdi et al., 2024). Regions with limited funding or infrastructure may struggle to adopt BNP-based systems without financial incentives or public–private partnerships. Furthermore, regulatory ambiguity surrounding nanoparticle discharge limits, lifecycle assessment, and environmental monitoring creates uncertainties that slow down commercialization and investment. Addressing these socio-economic factors through community-centered education, robust ecological policies, inclusive governance frameworks, and economically viable scale-up pipelines is crucial to ensuring that BNP-based wastewater treatment technologies achieve long-term societal acceptance and sustainable implementation (Rath et al., 2025a, b).

9 Regulatory Framework for Environmental Risk Mitigation

Most nations are still working to implement regulations, and the current regulatory frameworks for BNPs are dispersed across jurisdictions. The European Union has adopted a risk-based regulatory framework that requires manufacturers to demonstrate that BNPs do not pose significant environmental risks when production exceeds 1 ton per year (Bracking et al., 2021). Before using the particles commercially, manufacturers must determine their size distribution, surface chemistry, and aggregation behavior as part of essential management practices. Treatment facilities must maintain real-time monitoring systems that indicate threshold levels of 0.1–0.5 mg/l for total nanoparticle concentration in effluent discharges. Best-practice standards strongly encourage multiple containment barriers, including advanced filtration systems capable of capturing particles as small as 10 nm (Farka et al., 2024). Facilities must establish emergency action plans for accidental leaks and report any incidents involving concentrations exceeding 1 mg/L (Mäkkä et al., 2021). When handling concentrated BNP solutions,

worker safety guidelines require personal protective equipment and regular exposure evaluations. As part of environmental monitoring programs, it is crucial to conduct quarterly assessments of soil and water bodies within a 5 km radius of treatment facilities (Kumar et al., 2024).

10 Conclusion and Future Perspectives

The current global scenario is heavily influenced by the availability of safe, clean water, which is significantly affected by factors such as climate change and the release of toxic substances into water sources. BNPs are a promising class of environmentally compatible nanomaterials with significant potential to advance wastewater treatment processes. Unlike conventional nanoparticles, plant-, bacterial-, fungal-, algal-, and yeast-derived BNPs possess biomolecule-rich surfaces that confer enhanced catalytic activity, selective adsorption, redox reactivity, and antimicrobial activity, making them particularly effective for the removal of heavy metals, dyes, organic pollutants, pharmaceuticals, and pathogenic microorganisms. This review shows that the sustainability of BNPs arises not only from their biological origin but also from their reduced chemical toxicity, lower energy requirements during synthesis, and multifunctional mechanisms that outperform several traditional remediation techniques. The emerging literature indicates that BNPs can achieve high removal efficiencies across diverse wastewater systems, including industrial, municipal, agricultural, and hospital effluents, where limited treatment infrastructure and high pollutant loads necessitate cost-effective, scalable solutions. BNPs offer a feasible pathway for decentralized, adaptable water treatment technologies in these regions, provided synthesis consistency and regulatory frameworks are strengthened. International trends, including recent developments in the nano-risk governance and nanotechnology safety initiatives, reflect a growing recognition of the need for responsible innovation in nano-enabled remediation strategies. However, the future deployment of BNPs must be guided by a clear understanding of their environmental fate. Post-treatment transformations, such as sulfidation, oxidation, dissolution, aggregation, and organometallic complex formation, significantly affect the mobility, persistence, and ecological impact of BNPs in soils, sediments, and surface waters. These processes require

transformation-aware risk assessments, improved bio-solid management protocols, and long-term monitoring frameworks to ensure environmental safety.

Future research should prioritize scalable synthesis platforms, cost-reduction strategies, nanoparticle recovery techniques, and real-world pilot studies reflect the chemical and biological complexities of wastewater systems. Advancing BNP-based technologies will also require harmonized international regulations, community acceptance, and collaborative initiatives among academia, industry, and government. Overall, BNPs offer substantial potential as sustainable nanotechnological tools for wastewater remediation. Still, their widespread adoption will depend on bridging existing knowledge gaps in environmental fate, economic feasibility, regulatory governance, and large-scale implementation. Strengthening these dimensions will enable BNPs to transition from laboratory innovation to practical, environmentally safe water treatment solutions.

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